

EE-312L Signals and Systems Lab
Laboratory Session 07
**To Understand and Demonstrate the
Convolution of LTI Systems Using MATLAB**

October 29, 2024

Outlines:

1. Objective
2. Linear Time-Invariant (LTI) Systems
3. Impulse Response
4. Convolution of LTI Systems
5. Use MATLAB to verify Linearity Property

1 Objective

The objective of this lab session is to understand and demonstrate the convolution of Linear Time-Invariant (LTI) systems using MATLAB.

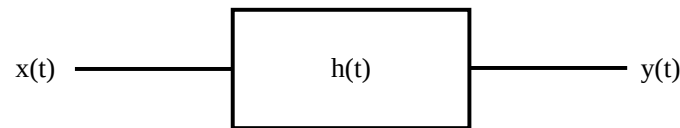
2 Linear Time-Invariant Systems

Linear time-invariant systems are a class of systems that are both linear and time-invariant. For continuous-time LTI system shown in Figure 1,

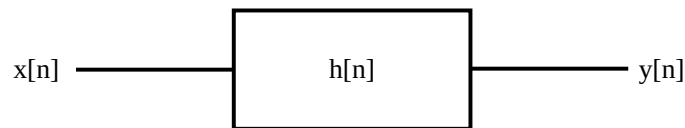
- $x(t)$ is the input (excitation) to the system
- $y(t)$ is the output (response) of the system
- $h(t)$ is the impulse response of the system

For discrete-time LTI system shown in Figure 1,

- $x[n]$ is the input (excitation) to the system
- $y[n]$ is the output (response) of the system
- $h[n]$ is the impulse response of the system



Continuous-Time LTI System



Discrete-Time LTI System

Figure 1: Linear Time-Invariant Systems

3 Impulse Response of System

In continuous-time LTI systems if the input is a unit impulse function to the system, i.e., $x(t) = \delta(t)$, then the response, $y(t) = h(t)$, as shown in Figure 2.

In discrete-time LTI systems if the input is a unit impulse function to the system, i.e., $x[n] = \delta[n]$, then the response, $y[n] = h[n]$, as shown in Figure 2.

4 Convolution of LTI Systems

Convolution is usually defined via the convolution integral for continuous time functions or the convolution sum for discrete-time functions.

4.1 Convolution Integral

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau \quad (1)$$

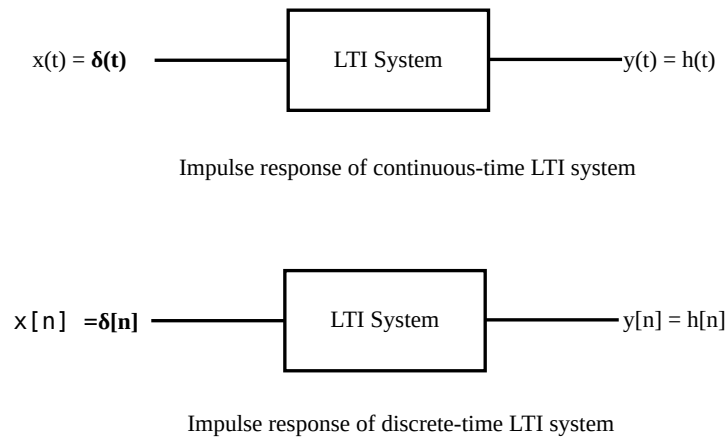


Figure 2: Impulse response of LTI system

4.2 Convolution Sum

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} x[n-k]h[k] = \quad (2)$$

5 Computing Convolution Integral using conv() [1]

- MATLAB provides a function `conv()` that performs a discrete-time convolution of two discrete-time sequences.
- Let make a new function `convint()` that uses `conv()` to numerically integrate the continuous-time convolution is as follows.

```

1 function [y,ty]=convint(x,tx,h,th)
2 %Inputs:
3 %x is the input signal vector
4 %tx is the times of the samples in x
5 %h is the impulse response vector
6 %th is times of the samples in h
7 %outputs:
8 %y is the output signal vector,
9 %length(y)=length(x)+length(h)-1
10 %ty is the time of the samples in y
11 dt=tx(2)-tx(1);
12 y=conv(x,h)*dt;
13 ty=(tx(1)+th(1))+[0:(length(y)-1)]*dt;
```

Example 1:

Consider the continuous-time convolution of $h(t)$ and $x(t)$, given as follow:

$$x(t) = \begin{cases} 1, & 0 \leq t \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

$$h(t) = \begin{cases} 1, & 0 \leq t \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

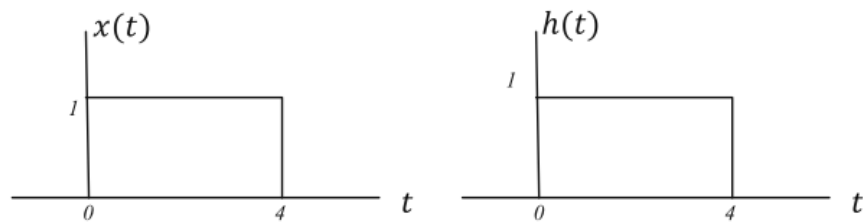


Figure 3: $x(t)$ and $h(t)$ for Example 1.

Use MATLAB to find convolution $y(t) = x(t) * h(t)$.

Here is the code

```

1  clc;
2  clear all;
3  close all;
4  tx=[0:0.01:4];
5  x=ones(1,length(tx));
6  th=[0:0.01:4];
7  h=ones(1,length(th));
8  [y ty]=convint(x,tx,h,th);
9
10 figure;
11 plot(ty,y, 'r', 'linewidth', 2.5);
12 xlabel('t')
13 ylabel('y(t)');
14 axis([0 8 0 4]);
15 grid minor on;
16 saveas(gcf, 'lab07plots/plot01.pdf')
```

6 Computing Convolution Sum using conv() [1]

The MATLAB function `conv(a,b)` can be used to compute convolution sum of two sequences.

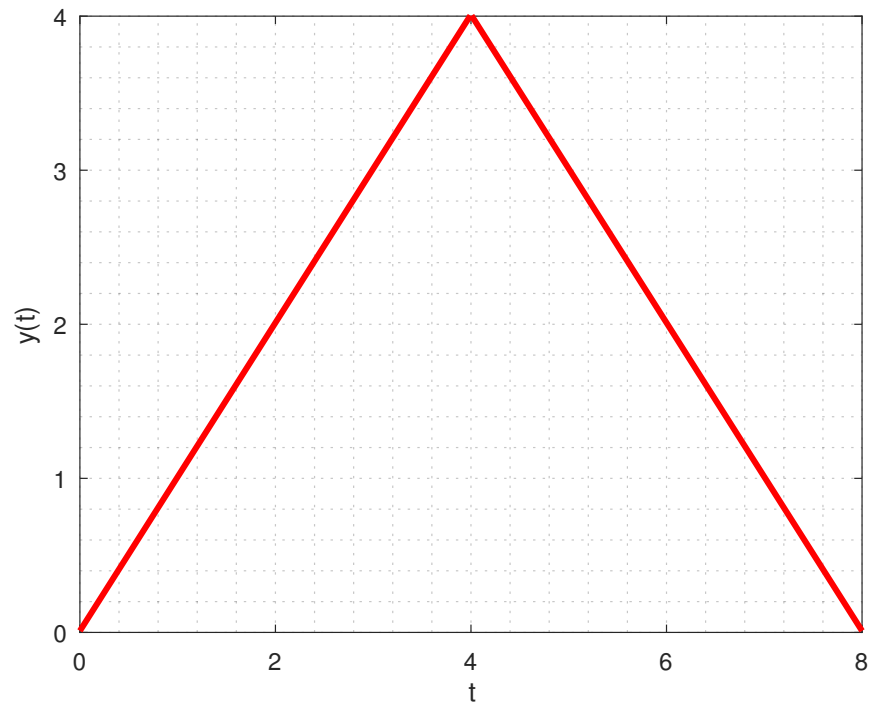


Figure 4: Convolution result of $y(t) = x(t) * h(t)$ for Example 1.

Example 2:

Compute convolution sum of the sequences $x[n] = \{2, -1, 0, 0\}$ and $h[n] = \{-1, 2, 1\}$ using MATLAB.

Here is the code

```
1 x=[ 2 -1 0 0 ];
2 h=[-1 2 1];
3 c=conv(x,h);
4 len=length(c)-1;
5 n=0:1:len;
6 stem(n,c, 'r', 'linewidth', 2.5)
7 xlabel('n');
8 ylabel('y[n]');
9 axis([0 5 -3 5]);
10 grid minor on;
11 saveas(gcf, 'lab07plots/plot02.pdf ' );
```

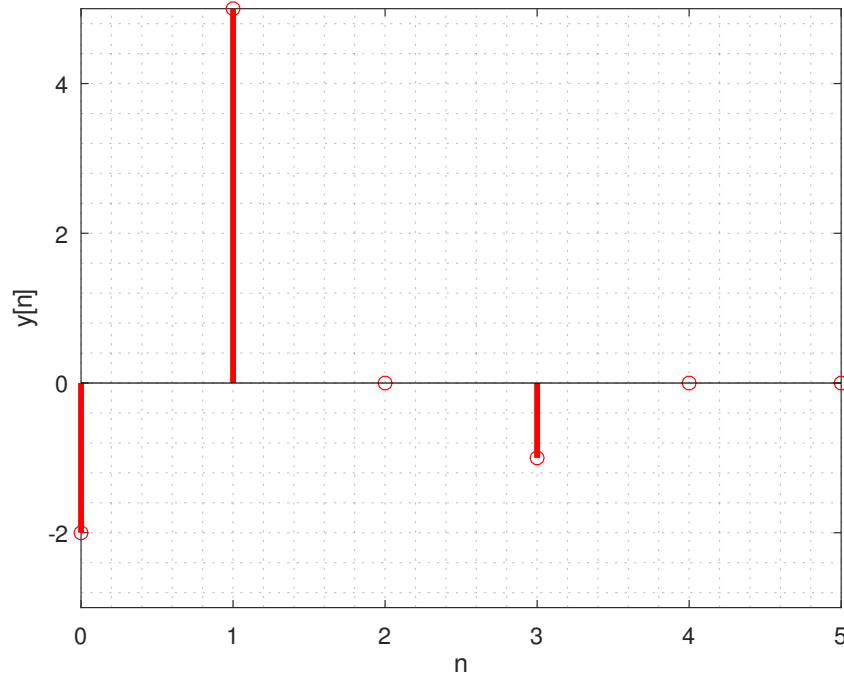


Figure 5: Convolution result of $y[n] = x[n] * h[n]$ for Example 2.

7 Computation of Impulse Response Using MATLAB

The impulse response of LTI discrete-time systems can be computed using MATLAB function:

$$y = \text{filter}(b, a, x),$$

where

- a is coefficient vector of input x
- b is coefficient vector of output y

Example 3:

Determine the impulse response of a discrete-time system described by the following difference equation:

$$y[n] - 2y[n - 1] = x[n] - 2x[n - 1] - 0.06x[n - 2] \quad (3)$$

In this equation,

- $a = [1, -2]$
- $b = [1, -2, 0.06]$

8 Lab 07: Lab Tasks

1. Write a MATLAB program to compute the convolution of the input $x(t)$ and the impulse response $h(t)$ shown in Figure 6.
2. Verify Commutative property of convolution i.e. $x(t) * h(t) = h(t) * x(t)$

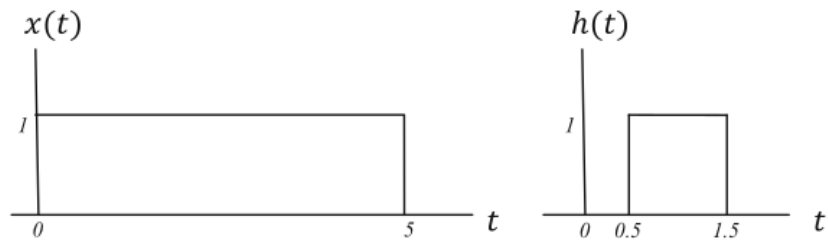


Figure 6: $x(t)$ and $h(t)$ for Lab Task.

References

- [1] K Deergha Rao and Satish Tunga. *Signals and systems*. Springer, 2018.